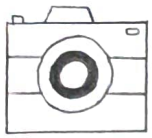
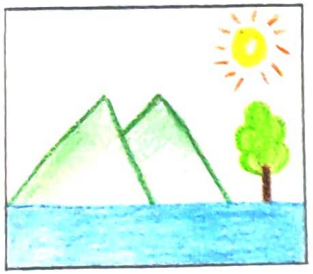
SMART PHOTO EDITING, OBJECT RECOGNITION, AR APP

INPUT
Real world image from a camera

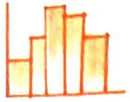




1. Image Processing

* Image enhancement
* Noise removal
* Filtering
* Edge detection
* color correction

Example: Brightness and blur correction



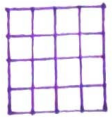


Image preprocessing prepares clearer data for vision tasks

2. Computer Vision

* Feature extraction
* Object detection
* Face / Landmark detection
* classification

Example: Detecting a face or a cat





3. Levels of Vision

Low level vision
Edges, filtering, noise removal

Mid level vision
Features, shapes, textures, corners

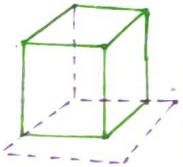
High level vision
Object recognition, classification, scene understanding

Guided virtual placement

4. Computer Graphics (AR Rendering)

* 2D / 3D rendering
* Virtual object placement
* Scaling, rotation, lighting
* Real-time overlay

Example: Overlaying virtual glasses on detected face



* AR try-on
* Filters
* Background replacement
* Smart editing
* ★ Real world examples

OUTPUT (AR VIEW)



5. Applications

Smart photo Enhancement



AR Try-on (glasses)



Background Replacement



Object recognition



Face filters & beauty apps



Interactive learning

Instant product Search

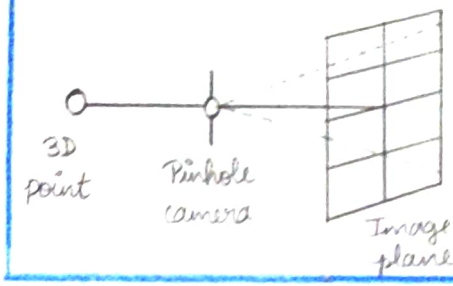
Legend (Links)

→ Data flow
→ Information flow
→ Control / Guides
→ Supports / Enhances
→ Final Output

AUTONOMOUS DRONE : 3D SCENE RECONSTRUCTION

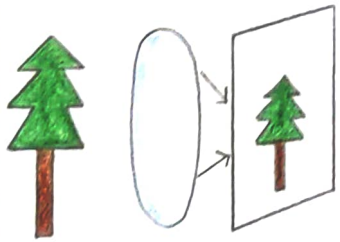
1. Camera model

* Pinhole camera model
* Focal length (f)
* Intrinsic parameters (K)
* Extrinsic parameters (R, t)
* 3D to 2D projection



2. Image formation

* Light enters the lens
* Scene projects onto image plane / sensor
* 3D scene becomes 2D image



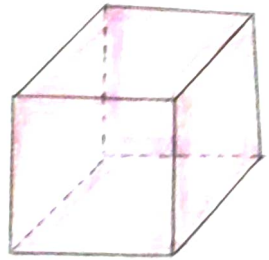
3. Camera calibration

* Estimate intrinsic parameters (f, K, distortion)
* Estimate lens distortion (radial, tangential)
* Estimate extrinsic parameters (camera pose)
* Rectification

5. 3D reconstruction (output)

* Combine depth, features & camera poses
* Reconstruct 3D structure
* Generate 3D point cloud / mesh / model

Output :
3D point cloud / 3D model

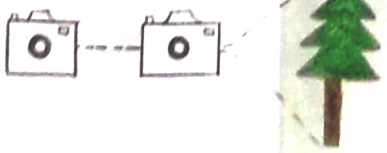


Flow of Data

Capture Image → Calibrate Cameras → Estimate depth → Reconstruct 3D Scene

4. A) Binocular Vision

* Two cameras (L/R)
* Find corresponding points
* Disparity calculation
* Depth estimation



4. B) Multiple View Geometry

* Multiple images from different viewpoints
* Feature matching across images
* Camera pose estimation
* 3D point estimation (triangulation)

Real World applications

* Mapping & Surveying
* Disaster management
* Agriculture monitoring
* 3D monitoring of environments

Key references

* Zhang, Z. (2000). A flexible new technique for camera calibration
* Hartley, R. & Zisserman, A. (2004). Multiple view geometry in computer vision